

Lab 1

Attendance



<https://forms.gle/6bdQxkKAHQiHPw176>

slides

https://blackboard.gwu.edu/ultra/courses/_446157_1/cl/outline

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Big-O Step-by-Step Proof Example 3 (Quadratic)

- Claim: Prove that $f(n) = 4n^2 + 7n + 6 = O(n^2)$.

Big-O Step-by-Step Proof Example 3 (Quadratic)

Goal: Find c, n_0 such that $4n^2 + 7n + 6 \leq c * n^2$ for all $n \geq n_0$.

Step-by-Step Proof:

1. Assume $n \geq 1$. Establish bounds for low-order terms relative to n^2 :

- $7n \leq 7n^2$

- $6 \leq 6n^2$

2. Sum the inequalities:

$$4n^2 + 7n + 6 \leq 4n^2 + 7n^2 + 6n^2 = 17n^2$$

3. Set $c = 17$ and $n_0 = 1$.

Check condition: $4(1)^2 + 7(1) + 6 = 17 \leq 17(1)^2$. Holds.

Big-O Step-by-Step Proof Example (Quadratic — Nontrivial n_0)

- Claim: Prove that $f(n) = 4n^2 + 7n + 6 = O(n^2)$.

Big-O Step-by-Step Proof Example (Quadratic — Nontrivial n_0)

- Claim: Prove that $f(n) = 4n^2 + 7n + 6 = O(n^2)$.
- Goal: Find c, n_0 such that $4n^2 + 7n + 6 \leq c \cdot n^2$ for all $n \geq n_0$.

Step-by-Step Proof:

1. Assume $n \geq 3$. Establish bounds for low-order terms relative to n^2 :
 - $7n \leq (7/3)n^2$
 - $6 \leq (2/3)n^2$
2. Sum the inequalities:
$$4n^2 + 7n + 6 \leq 4n^2 + (7/3)n^2 + (2/3)n^2 = 7n^2$$
3. Set $c = 7$ and $n_0 = 3$.

Check condition: $4(3)^2 + 7(3) + 6 = 63 \leq 7(3)^2 = 63$. Holds.

Takeaways

- n_0 is a cutoff point: “I don't care what happens before n_0 . Starting at n_0 , my function is always below cn^2 .”
- c and n_0 work together: $4n^2+7n+6=O(n^2)$: $(c,n_0)=(16,1)$ $(c,n_0)=(7,4)$
- You're proving an upper bound, not finding the exact value.
- Big-O proof = Find ANY $c>0$ and $n_0>0$ such that $f(n)\leq c g(n)$ for EVERY $n\geq n_0$.

Big-Omega Step-by-Step Proof Example (Quadratic)

- Claim: Prove that $f(n) = (3/4)n^2 - 6n = \Omega(n^2)$.

Big-Omega Step-by-Step Proof Example (Quadratic)

- Claim: Prove that $f(n) = (3/4)n^2 - 6n = \Omega(n^2)$.
- Goal: Find c, n_0 such that $(3/4)n^2 - 6n \geq c * n^2$ for all $n \geq n_0$.

Assume $n \geq 24$. Establish a lower bound for the negative lower-order term: $\circ 6n \leq (1/4)n^2$

$$\circ -6n \geq -(1/4)n^2$$

Substitute this bound into $f(n)$: $(3/4)n^2 - 6n \geq (3/4)n^2 - (1/4)n^2 = (1/2)n^2$

3. Set $c = 1/2$ and $n_0 = 24$. Check condition:

$$(3/4)(24)^2 - 6(24) = 432 - 144 = 288$$

$$(1/2)(24)^2 = 288.$$

Thus, $288 \geq 288$. Holds.

Big-Theta Step-by-Step Proof Example (Quadratic)

- Claim: Prove that $f(n) = 5n^2 + 8n + 3 = \Theta(n^2)$.

Big-Theta Step-by-Step Proof Example (Quadratic)

- Claim: Prove that $f(n) = 5n^2 + 8n + 3 = \Theta(n^2)$.
- Goal: Find c_1 , c_2 , n_0 such that

$c_1n^2 \leq 5n^2 + 8n + 3 \leq c_2n^2$ for all $n \geq n_0$.

1. Lower bound: Assume $n \geq 1$.
 $5n^2 + 8n + 3 \geq 5n^2$
Set $c_1 = 5$.
2. Upper bound: Since $n \geq 1$:
 $8n \leq 8n^2$, $3 \leq 3n^2$
 $5n^2 + 8n + 3 \leq 5n^2 + 8n^2 + 3n^2 = 16n^2$
Set $c_2 = 16$.
3. Set $n_0 = 1$.
4. Combine:
 $5n^2 \leq 5n^2 + 8n + 3 \leq 16n^2$ for all $n \geq 1$.

Conclusion: $f(n) = 5n^2 + 8n + 3 = \Theta(n^2)$ [Q.E.D.]